

Physical Chemistry of Materials

Lecture 07

Interacting Particles

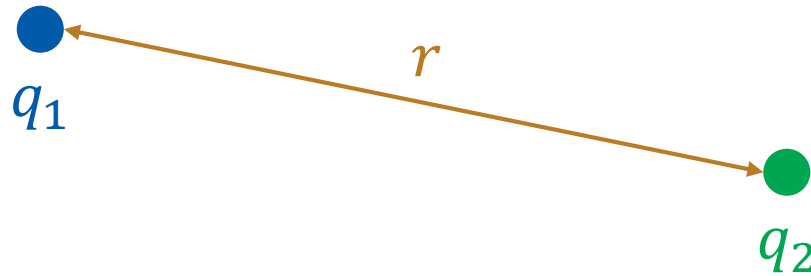


Intermolecular Interactions

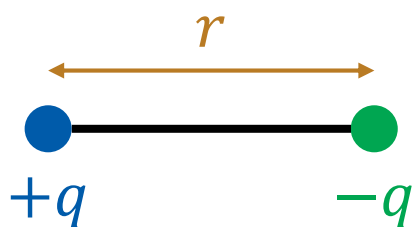
- Charge-charge interactions
- Charge-dipole interactions
- Dipole-dipole interactions
- Dispersion force
- Pauli repulsion
- Hydrogen bonds

Charge-Charge Interactions

$$V(r) = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$$



Electric Dipoles



Dipole moment

$$\vec{\mu} = q\vec{r}$$

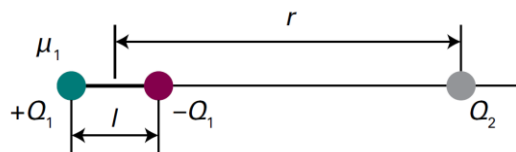
Units: Debye
 $1 \text{ D} = 3.34 \times 10^{-30} \text{ C m}$

For polyatomic molecules,

$$\vec{\mu} = \sum_i q_i \vec{r}_i$$

Molecule	Dipole moment
CCl_4	0
H_2	0
H_2O	1.85 D
HCl	1.08 D
HI	0.42 D

Charge-Dipole Interactions



$$V(r) = \frac{1}{4\pi\epsilon_0} \left(-\frac{Q_1 Q_2}{r - l/2} + \frac{Q_1 Q_2}{r + l/2} \right)$$

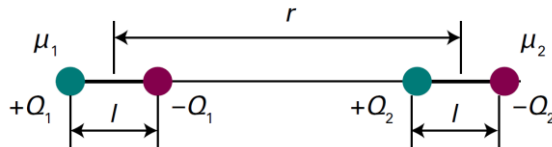
Letting $x = l/2r$,

$$V(r) = \frac{Q_1 Q_2}{4\pi\epsilon_0 r} \left(-\frac{1}{1 - x} + \frac{1}{1 + x} \right) \overset{x \ll 1 \text{ (or } l \ll r)}{\approx} -\frac{Q_1 Q_2}{4\pi\epsilon_0 r} \times 2x = -\frac{Q_1 Q_2 l}{4\pi\epsilon_0 r^2}$$

Hence,

$$V(r) = -\frac{\mu_1 Q_2}{4\pi\epsilon_0 r^2}$$

Dipole-Dipole Interactions



$$V(r) = \frac{1}{4\pi\epsilon_0} \left(-\frac{Q_1 Q_2}{r+l} + \frac{Q_1 Q_2}{r} + \frac{Q_1 Q_2}{r} - \frac{Q_1 Q_2}{r-l} \right)$$

Letting $x = l/r$,

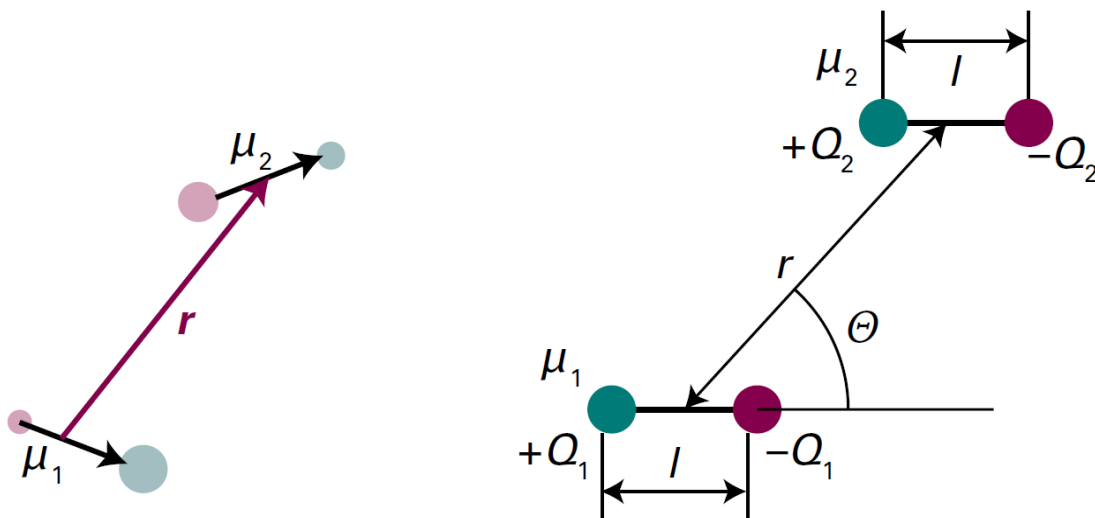
$$V(r) = -\frac{Q_1 Q_2}{4\pi\epsilon_0 r} \left(\frac{1}{1+x} - 2 + \frac{1}{1-x} \right) \overset{x \ll 1 \text{ (or } l \ll r)}{\approx} -\frac{Q_1 Q_2}{4\pi\epsilon_0 r} \times 2x^2 = -\frac{2Q_1 Q_2 l^2}{4\pi\epsilon_0 r^3}$$

Hence,

$$V(r) = -\frac{2\mu_1 \mu_2}{4\pi\epsilon_0 r^3}$$

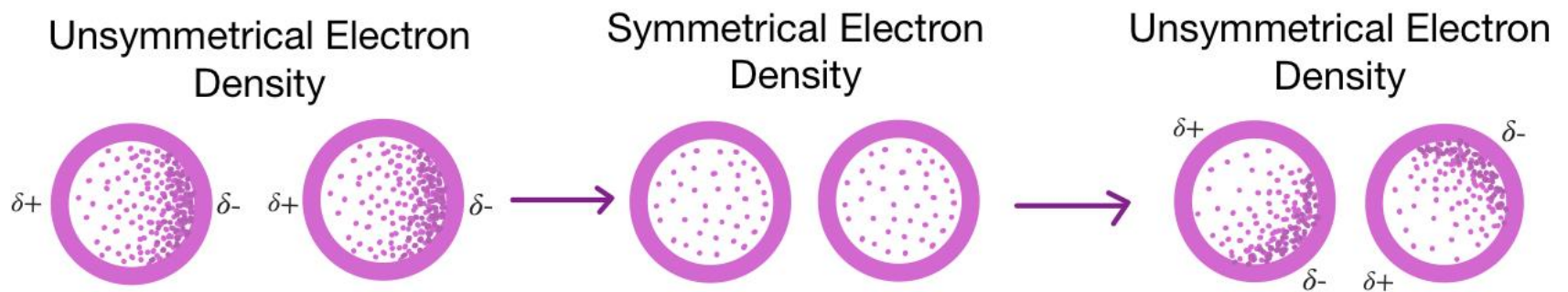
(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th Ed.)

In General



$$V(r) = \frac{\mu_1 \mu_2}{4\pi\epsilon_0 r^3} (1 - 3 \cos^2 \Theta)$$

Dispersion Force



Induced dipole-induced dipole interaction

Two Hydrogen Atoms

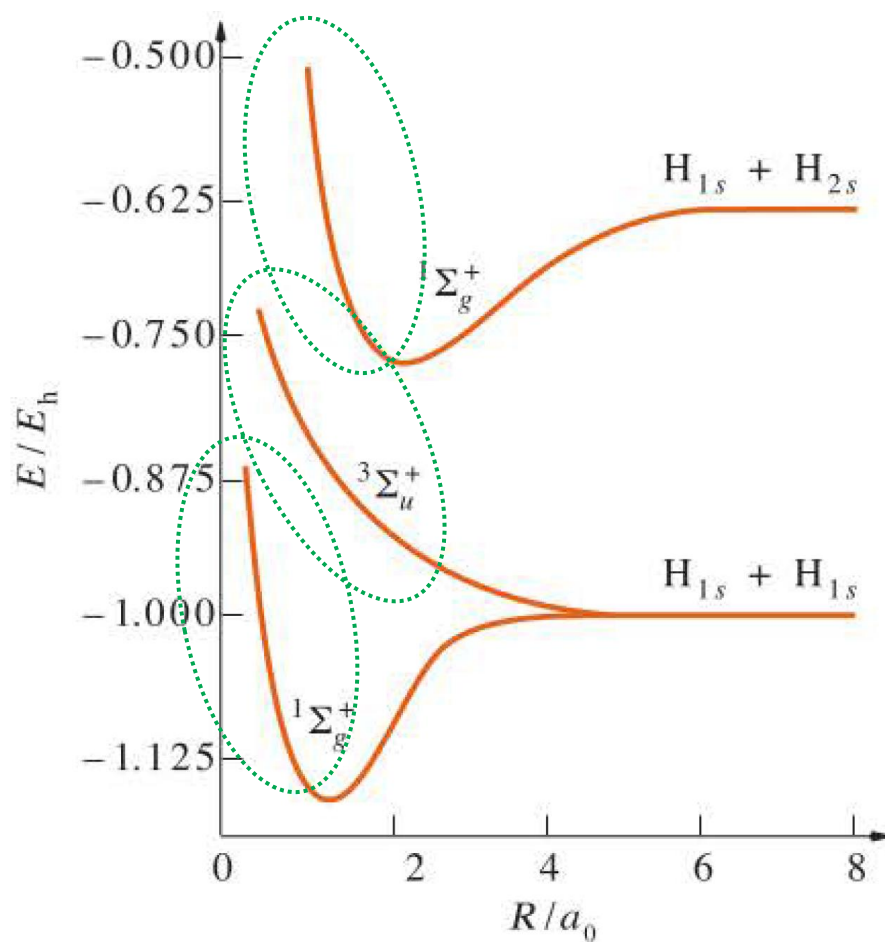


- Hamiltonian $\hat{H} = \underbrace{\hat{H}_A + \hat{H}_B}_{\hat{H}^{(0)}} + \underbrace{\hat{H}_{\text{dip-dip}}}_{\hat{H}^{(1)}}$ $\hat{H}_{\text{dip-dip}} \propto \frac{\mu_1 \mu_2}{r^3}$
- Perturbation theory

$$E^{(1)} = 0$$

$$E^{(2)} \propto -\hat{H}_{\text{dip-dip}}^2 = -\frac{1}{r^6}$$

Pauli Repulsion



(Image: McQuarrie, *Quantum Chemistry*, 2nd Ed., Figure 11.12)

Approximations of Pauli Repulsion

- Hard sphere potential

$$V(r) = \begin{cases} \infty, & r < \sigma \\ 0, & r \geq \sigma \end{cases}$$

- Power-law potential (n : integer between 9 and 16)

$$V(r) = \left(\frac{\sigma}{r}\right)^n$$

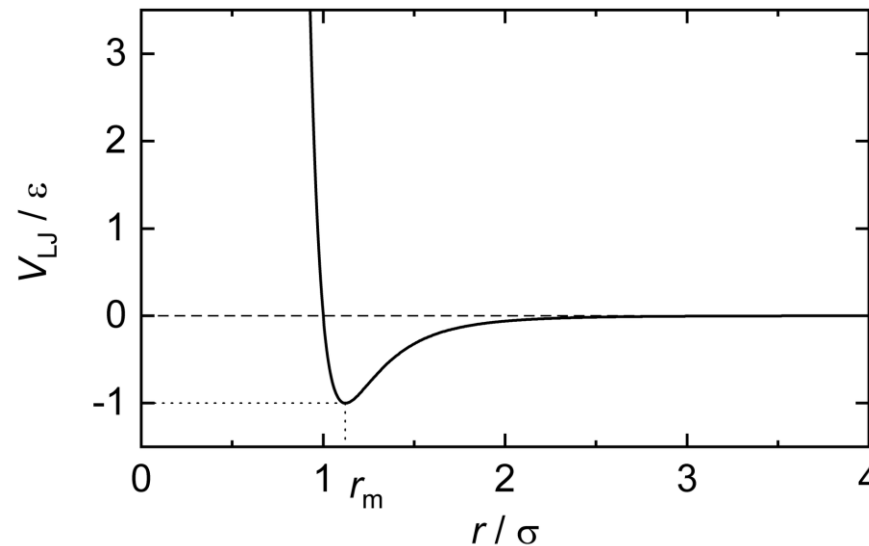
- Exponential potential

$$V(r) = ce^{-r/\sigma_0}$$

Lennard-Jones Potential

Pauli repulsion + attractive dispersion force

$$V(r) = \frac{A}{r^{12}} - \frac{B}{r^6} = 4\varepsilon \left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^6 \right]$$



(Image: https://en.wikipedia.org/wiki/File:Graph_of_Lennard-Jones_potential.png)

Potential Energy Minimum

$$V(r) = 4\varepsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right]$$

- Equilibrium distance r_m

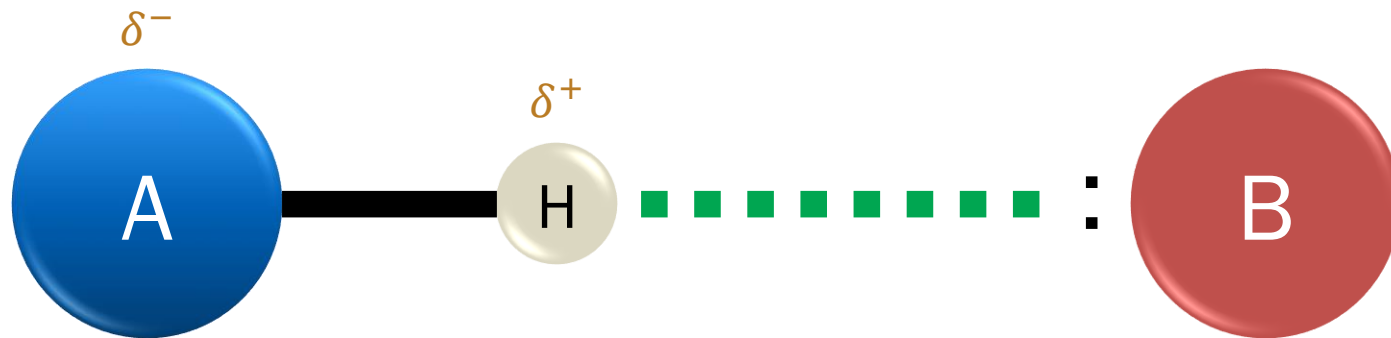
$$\frac{dV}{dr} = 4\varepsilon \left(-12 \frac{\sigma^{12}}{r^{13}} + 6 \frac{\sigma^6}{r^7} \right) = 0$$

$$r_m^6 = 2\sigma^6 \text{ or } r_m = 2^{1/6}\sigma$$

- Potential energy

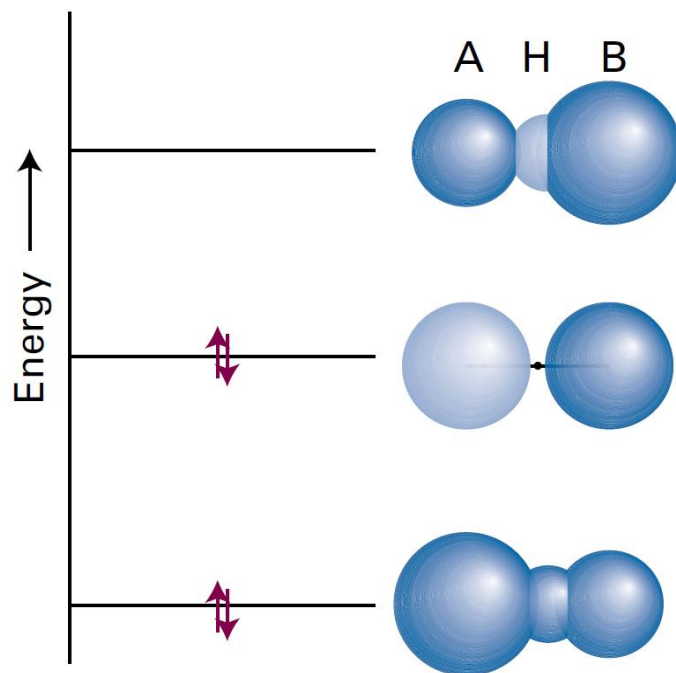
$$V(r_m) = 4\varepsilon \left(\frac{1}{4} - \frac{1}{2} \right) = -\varepsilon$$

Hydrogen Bonds



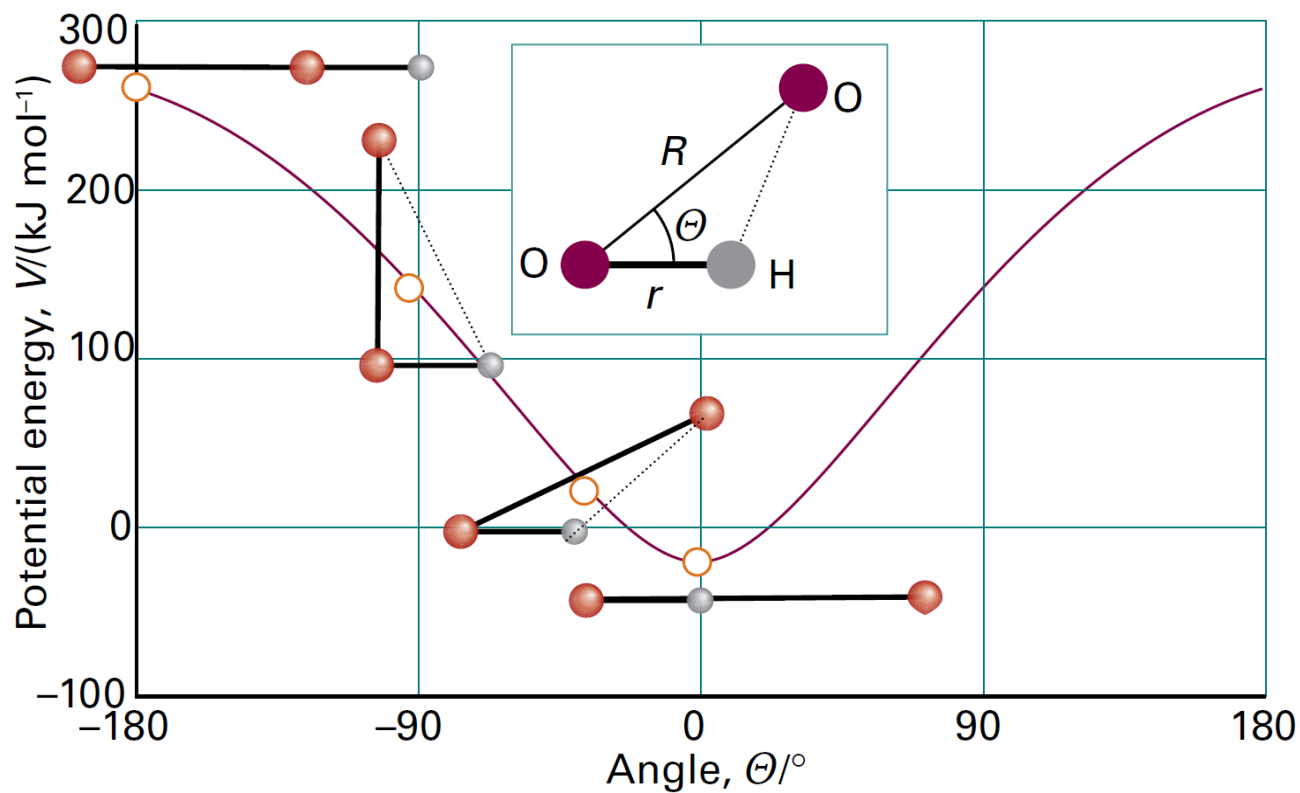
Dipole-dipole interaction?

Hydrogen Bonds: MO Picture



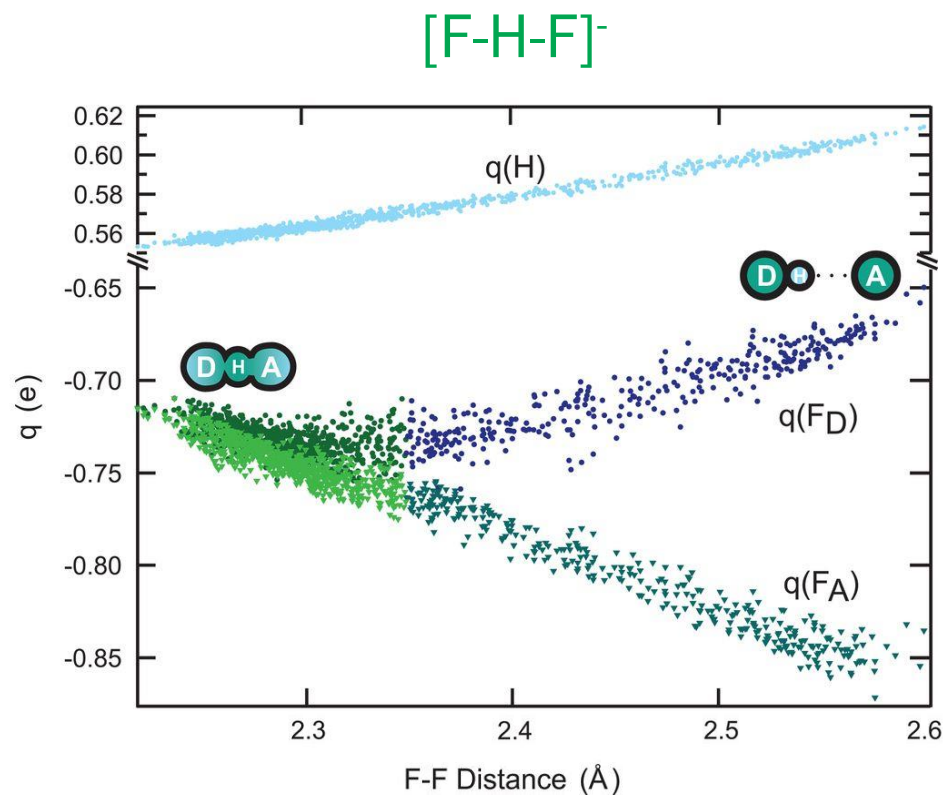
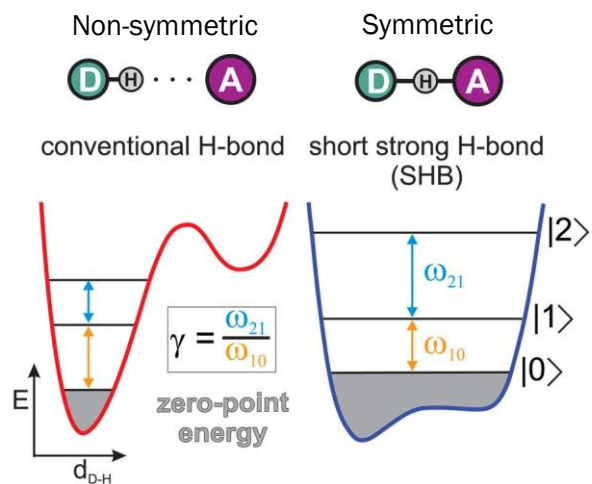
$$\phi = c_1\psi_A + c_2\psi_H + c_3\psi_B$$

Orientation Dependence



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th Ed., Fig. 14B.6)

Covalent Bond or Dipole-Dipole?



(Image: Dereka et al., *Science*, 2021)

Summary

Interaction type	Distance dependence of potential energy	Typical energy/ (kJ mol ⁻¹)	Comment
Ion-ion	$1/r$	250	Only between ions
Hydrogen bond		20	Occurs in X-H...Y, where X, Y = N, O, or F
Ion-dipole	$1/r^2$	15	
Dipole-dipole	$1/r^3$	2	Between stationary polar molecules
	$1/r^6$	0.3	Between rotating polar molecules
London (dispersion)	$1/r^6$	2	Between all types of molecules and ions

(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th Ed., Table 14B.1)



Interactions and Phase Transitions

- Phase transitions: **ordered** $\leftrightarrow$ **disordered**
- e.g. liquid-gas transition, magnetic transition
- Competition between **energy** and **entropy**
- **Interactions** are required!

(Monatomic) Gas Models

- Ideal gas

- Energy $E = \sum_i (p_i^2 / 2m)$
- Partition function

$$Z = \frac{1}{N! h^{3N}} \iint e^{-\sum_i p_i^2 / 2mk_B T} d^N \vec{r} d^N \vec{p} = \frac{V}{N!} \left(\frac{2\pi m k_B T}{h^2} \right)^{3N/2}$$

- Real gas

- Energy $E = \sum_i (p_i^2 / 2m) + \sum_{i < j} E_{\text{pot}}(r_{ij})$

Partition Function

$$E = \sum_i (p_i^2 / 2m) + \sum_{i < j} E_{\text{pot}}(r_{ij})$$

$$Z = \frac{1}{N! h^{3N}} \left(\frac{2\pi m k_B T}{h^2} \right)^{3N/2} \int e^{-\sum_{i < j} E_{\text{pot}}(r_{ij}) / k_B T} d^N \vec{r}$$

Now define

$$Z_{\text{pot}} = \int e^{-\sum_{i < j} E_{\text{pot}}(r_{ij}) / k_B T} d^N \vec{r}$$

Mayer f -Function

$$f_{ij} = e^{-E_{\text{pot}}(r_{ij})/k_{\text{B}}T} - 1$$

$$\begin{aligned} Z_{\text{pot}} &= \int e^{-\sum_{i<j} E_{\text{pot}}(r_{ij})/k_{\text{B}}T} d^N \vec{r} \\ &= \int \left[\prod_{i<j} e^{-E_{\text{pot}}(r_{ij})/k_{\text{B}}T} \right] d^N \vec{r} \\ &= \int \left[\prod_{i<j} (1 + f_{ij}) \right] d^N \vec{r} \end{aligned}$$

Mayer f -Function

$$Z_{\text{pot}} = \int \left[\prod_{i < j} (1 + f_{ij}) \right] d^N \vec{r}$$

$$Z_{\text{pot}} = \int \left(1 + \sum_{i < j} f_{ij} + \dots \right) d^N \vec{r}$$

Assuming $f_{ij} \ll 1$ (i.e., $E_{\text{pot}}(r_{ij}) \ll k_B T$),

$$Z_{\text{pot}} \approx \int \left(1 + \sum_{i < j} f_{ij} \right) d^N \vec{r} = V^N + \int \sum_{i < j} f_{ij} d^N \vec{r}$$

Mayer f -Function

$$Z_{\text{pot}} = V^N + \int \sum_{i < j} f_{ij} d^N \vec{r}$$

$$Z_{\text{pot}} = V^N + V^{N-1} \frac{N(N-1)}{2} \int \left(e^{-E_{\text{pot}}(r)/k_{\text{B}}T} - 1 \right) d\vec{r}$$

$$= I$$

$$Z_{\text{pot}} \approx V^N \left(1 + \frac{N^2}{2V} I \right)$$

Helmholtz Energy and Pressure

- Helmholtz energy $A = -k_B T \ln Z$
- Pressure

$$\begin{aligned} p &= - \left(\frac{\partial A}{\partial V} \right)_{T,n} = k_B T \left(\frac{\partial \ln Z}{\partial V} \right)_{T,n} = k_B T \left(\frac{\partial \ln Z_{\text{pot}}}{\partial V} \right)_{T,n} \\ &= k_B T \frac{\partial}{\partial V} \left[N \ln V + \ln \left(1 + \frac{N^2 I}{2V} \right) \right] \\ &\approx k_B T \left(\frac{N}{V} - \frac{N^2 I}{2V^2} \right) \end{aligned}$$

Evaluation of I

$$I = \int (e^{-E_{\text{pot}}(r)/k_{\text{B}}T} - 1) d\vec{r} = \int_0^\infty (e^{-E_{\text{pot}}(r)/k_{\text{B}}T} - 1) 4\pi r^2 dr$$

Need to specify $E_{\text{pot}}(r)$

- **Sutherland potential:** hard sphere + dispersion force

$$E_{\text{pot}}(r) = \begin{cases} +\infty, & r < r_0 \\ -\epsilon_0 \left(\frac{r_0}{r}\right)^6, & r \geq r_0 \end{cases}$$

Two Regions

$$I = \int_0^{\infty} (e^{-E_{\text{pot}}(r)/k_{\text{B}}T} - 1) 4\pi r^2 dr$$

$$E_{\text{pot}}(r) = \begin{cases} +\infty, & r < r_0 \\ -\epsilon_0 \left(\frac{r_0}{r}\right)^6, & r \geq r_0 \end{cases}$$

$$I = \int_0^{r_0} (e^{-\infty/k_{\text{B}}T} - 1) 4\pi r^2 dr + \int_{r_0}^{\infty} (e^{\beta\epsilon_0(r_0/r)^6} - 1) 4\pi r^2 dr$$

Final Result

$$I = -\frac{4}{3}\pi r_0^3 + \int_{r_0}^{\infty} \left(e^{\frac{\epsilon_0}{k_B T} \left(\frac{r_0}{r} \right)^6} - 1 \right) 4\pi r^2 dr$$

Assuming $\epsilon_0 \ll k_B T$,

$$e^{\frac{\epsilon_0}{k_B T} \left(\frac{r_0}{r} \right)^6} \approx 1 + \frac{\epsilon_0}{k_B T} \left(\frac{r_0}{r} \right)^6$$

$$I = -\frac{4}{3}\pi r_0^3 + \int_{r_0}^{\infty} \frac{\epsilon_0}{k_B T} \left(\frac{r_0}{r} \right)^6 4\pi r^2 dr = -\frac{4}{3}\pi r_0^3 \left(1 - \frac{\epsilon_0}{k_B T} \right)$$

Pressure

$$p = k_B T \left(\frac{N}{V} - \frac{N^2 I}{2V^2} \right), \quad I = -\frac{4}{3} \pi r_0^3 \left(1 - \frac{\epsilon_0}{k_B T} \right)$$

Substitution leads to

$$p = \frac{Nk_B T}{V} + \frac{N^2 k_B T}{V^2} \cdot \frac{2}{3} \pi r_0^3 - \frac{N^2}{V^2} \cdot \frac{2}{3} \pi r_0^3 \epsilon_0$$

$$p + \frac{N^2}{V^2} \cdot \frac{2}{3} \pi r_0^3 \epsilon_0 = \frac{Nk_B T}{V} \left(1 + \frac{N}{V} \cdot \frac{2}{3} \pi r_0^3 \right)$$

$\underbrace{\hspace{10em}}$ $\underbrace{\hspace{10em}}$

a b

Equation of State

$$p + a \frac{N^2}{V^2} = \frac{Nk_{\text{B}}T}{V} \left(1 + b \frac{N}{V} \right)$$

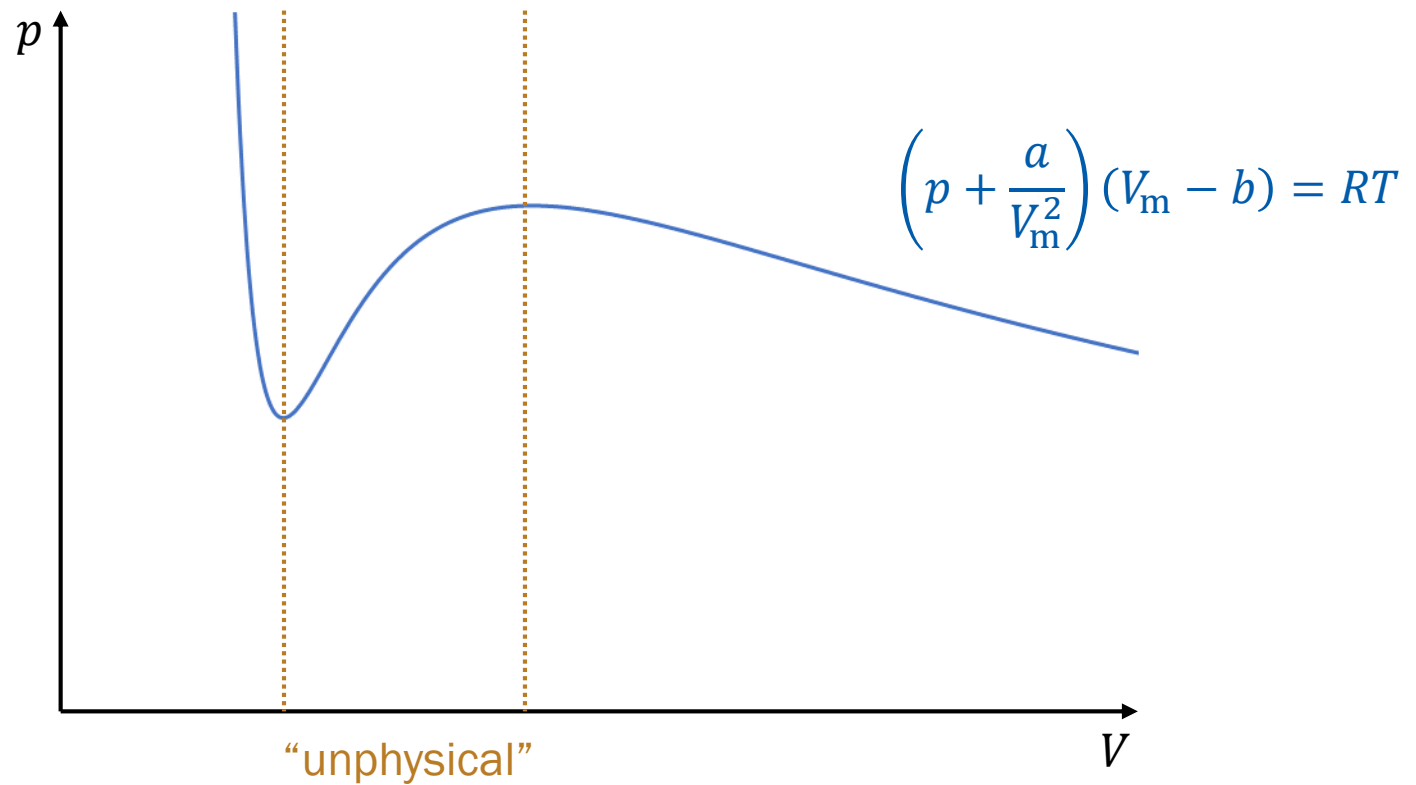
As $Nb/V \ll 1$,

$$\frac{Nk_{\text{B}}T}{V} = \frac{p + aN^2/V^2}{1 + bN/V} \approx \left(p + \frac{aN^2}{V^2} \right) \left(1 - \frac{bN}{V} \right)$$

$$\left(p + \frac{a}{V_{\text{m}}^2} \right) (V_{\text{m}} - b) = RT$$

(van der Waals equation of state)

vdW Equation (at low T)

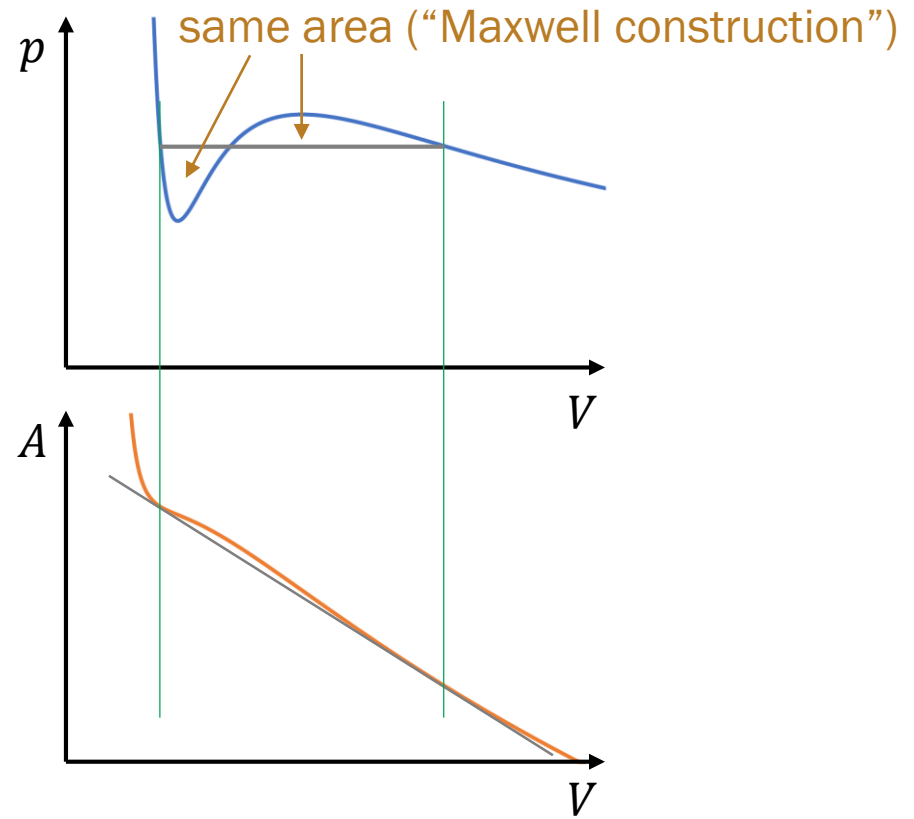


Metastable Phases

$$p = - \left(\frac{\partial A}{\partial V} \right)_{T,n}$$

At constant T and n ,

$$A = - \int p \, dV$$



Phase Transition!

